Randy Pahang II

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Education

University of Nevada, Las Vegas Expected May 2027
B.S. Computer Science *— Honors College — Wolzinger Family Engineering Scholar* *GPA: 4.0/4.0*

Experience

JT4 (Amentum) May 2026 – Present
Software Engineering Intern *Las Vegas, NV*
- Built a **C++23 telemetry simulator** sustaining **20 Mbps** of real-time flight-test data with precise packet timing, replacing commercial software and saving **$10K+ in licensing costs**.
- Designed a configurable packet-generation pipeline reproducing **flight-representative timing and data rates** to validate downstream telemetry processing before flight-test and range operations.
- Exposed the C++ simulation engine through a **C ABI** to a Python/PySide6 dashboard for real-time inspection of **100+ telemetry channels**.
- Validated generated telemetry against recorded flight-test data and built a GoogleTest suite with **93% coverage** across packet serialization, timing, and protocol edge cases.

STARS Solution LLC Nov 2025 – Jun 2026
Software Engineering Intern (Part-Time) *Las Vegas, NV*
- Designed and built **Java Spring Boot** backend services integrating institution data from **30,000+ schools** for a **3,500+ user pilot**.
- Reduced AI recommendation time-to-first-response from 8.2s to 0.4s by decoupling generation with **RabbitMQ** and streaming partial results via **Server-Sent Events**.
- Reduced institution-search latency from 110ms to 37ms using a **Redis cache-aside** layer with versioned invalidation and database fallback.
- Built **production observability** for Spring Boot AI workloads using **Micrometer and AWS CloudWatch**, with automated **SNS alerts** for provider failures, token exhaustion, and failed jobs.

Leadership

Bit2Byte Nov 2025 – Present
Co-Founder & Lead Developer *Las Vegas, NV*
- Co-founded UNLV's software development organization (100+ members), leading workshops, sprint planning, and code reviews for student teams building with **React/Next.js, Spring Boot, and FastAPI**.

Projects

PortRN | *Next.js, TypeScript, Spring Boot, PostgreSQL, RabbitMQ, AWS ECS, Stripe* Nov 2025 – Apr 2026
- Built and launched a full-stack AI portfolio and networking platform (**Next.js/TypeScript** frontend, **Spring Boot** API) on **AWS ECS** for a 50+ user beta, with idempotent **Stripe** billing.
- Parallelized multi-step AI generation across asynchronous **RabbitMQ workers** with section-level validation and repair of LLM output, reducing end-to-end processing from 7–10 to 4–5 minutes.
- Built a **skill-verification engine** that checks resume claims against GitHub commits, discounting forked or copied code so scores reflect each user's own work.
- Replaced unreliable iframe rendering of AI-generated React sections by **transpiling them in-browser with Babel** against a fixed library scope, with cached compiled output.

1st of 100+ Participants, RebelHacks 2026 | *HotelGuard* Jan 2026
- Built a concurrent pipeline using **YOLOv8 tracking** and parallel LLMs to generate room-level risk scores.

TrafficControl | *Go, TCP/UDP, Linux, YAML* Sep 2026 – Present
- Built a **Go** network traffic generator that drives concurrent **TCP/UDP** streams from declarative **YAML** configs, coordinated with goroutines and channels, sustaining **150K UDP packets/s**.
- Kept send rates exact under OS scheduling delays with a deadline-based pacer using **fractional-nanosecond carry** and catch-up bounded to a **5 ms** window, achieving **0.0007% median pacing error** across five 10s loopback runs.

Technical Skills

Languages: C/C++, Java, Python, Go, TypeScript, SQL
Frameworks & Data: Spring Boot, FastAPI, Node.js, React, Next.js, PostgreSQL, Redis, RabbitMQ
Cloud & Systems: AWS (ECS, EC2, S3), Docker, Linux, Git, CI/CD, CMake, GoogleTest